

4.1 Analytical Question: Critiquing PCA through the Lens of Algorithmic Fairness

Problem statement (unchanged).

Classical PCA's objective is to minimize the global average reconstruction error. Citing recent research directions, explain how this global objective can lead to unfair outcomes when the data contains distinct demographic groups. Define what "fairness" might mean in the context of dimensionality reduction, and describe the core idea behind one approach to "Fair PCA" (e.g., minimizing the maximum reconstruction loss across groups).

Solution (reformatted & clarified)

Classical PCA is optimal for minimizing the *global* mean squared reconstruction error, but that global objective can mask disparate performance across subpopulations. When the dataset contains multiple demographic groups with different covariance structures or variances, a single subspace chosen to minimize the overall average error tends to align with the dominant (high-variance or majority) group. As a result, minority groups can suffer higher reconstruction error, producing representations that are less faithful and potentially leading to downstream unfairness.

How the global objective induces unfairness

- PCA minimizes the average (over all points) reconstruction error:

$$\min_{W \text{ (rank-} r \text{)}} \mathbb{E}_{x \sim \text{data}} \|x - \hat{x}_W\|^2,$$

where $\hat{x}_W$ is the projection/reconstruction of x onto the rank- r subspace W .

- If group G_1 contributes most of the dataset variance or the majority of samples, the optimizer will prefer reducing error for G_1 even if that increases error for G_2 .
- Thus, average-optimality can hide large per-group disparities.

Possible definitions of "fairness" in dimensionality reduction

- **Equitable reconstruction error:** per-group reconstruction errors should be comparable (e.g., $\text{MSE}_k \approx \text{MSE}_{k'}$ for all groups k, k').
- **Parity of explained variance:** the proportion of variance explained per group should be similar.
- **Obfuscation of sensitive attribute:** the low-dimensional representation should make it difficult to infer the sensitive attribute.
- **Distributional parity:** projected distributions across groups should be close (e.g., via MMD).

Each definition suits different downstream goals; choosing one is an explicit normative modeling decision.

One approach: Minimax / group-robust PCA

Replace the average objective with a worst-group objective:

$$\min_W \max_{k \in \{1, \dots, K\}} \text{MSE}(W; X_k),$$

where $\text{MSE}(W; X_k)$ is the mean squared reconstruction error for group k . This produces a compromise subspace that bounds the worst-case reconstruction loss across groups. Practically, solving this minimax problem can be done via convex relaxations, iterative reweighting, or projected gradient methods on suitable parameterizations of W .

Trade-off: The minimax/fair solution typically increases global MSE (the “price of fairness”) but reduces disparity; practitioners must balance statistical efficiency and fairness goals.

4.2 Numerical Problem: Quantifying Representational Bias

Given:

- PCA from 10D $\rightarrow$ 2D.
- Group A: total variance $V_A = 50$, projected (captured) variance $P_A = 45$.
- Group B: total variance $V_B = 30$, projected variance $P_B = 18$.

(a) Reconstruction error (per group)

Use

$$\text{ReconstructionError} = \text{TotalVariance} - \text{ProjectedVariance}.$$

- Group A: $E_A = V_A - P_A = 50 - 45 = 5$.
- Group B: $E_B = V_B - P_B = 30 - 18 = 12$.

(b) Percentage of variance explained (per group)

$$\text{Prop}_k = \frac{P_k}{V_k}.$$

- Group A: $\text{Prop}_A = 45/50 = 0.90 = 90\%$.
- Group B: $\text{Prop}_B = 18/30 = 0.60 = 60\%$.

(c) Representational bias (difference)

$$\text{Bias} = \text{Prop}_A - \text{Prop}_B = 0.90 - 0.60 = 0.30 = 30 \text{ percentage points}.$$

This 30-point gap is large: Group A’s low-dim representation is much more faithful than Group B’s.

(d) Quick numeric check of global explained variance

For reference, the classical (global) percentage of variance explained (across both groups together) is

$$\frac{P_A + P_B}{V_A + V_B} = \frac{45 + 18}{50 + 30} = \frac{63}{80} = 0.7875 = 78.75\%.$$

If instead you average *normalized* explained variances equally (i.e., $0.5 \cdot 90\% + 0.5 \cdot 60\% = 75\%$), you see the “price of fairness” numerically: equal-weight normalized average is 75% vs classical overall 78.75%.

(e) Simple modification to mitigate bias

A straightforward, interpretable modification is to optimize a weighted objective that normalizes per-group contribution by group total variance (so small-variance groups don’t get drowned out), e.g.:

Normalized weighted objective

$$\max_W \sum_{k=1}^K w_k \cdot \frac{P_k(W)}{V_k},$$

where $P_k(W)$ is the variance captured for group k by subspace W , V_k is the total variance of group k , and weights $w_k \geq 0$ sum to 1.

- Choosing $w_k = 1/K$ yields equal emphasis on per-group relative explained variance (enforces parity tendency).
- Choosing w_k proportional to group importance lets you trade fairness vs efficiency.

Alternative / related formulations

- **Minimax (robust) objective:** $\min_W \max_k \text{MSE}(W; X_k)$ (directly controls worst-case reconstruction).
- **Constrained formulation:** maximize global explained variance subject to $\text{Prop}_k \geq \tau$ for each k (enforce minimum per-group fidelity).
- **Regularized objective (penalize disparity):**

$$\max_W (\text{global proj. variance}) - \lambda \cdot \text{Disparity}(\{\text{Prop}_k\}),$$

where Disparity could be $\max_k \text{Prop}_k - \min_k \text{Prop}_k$ or $\sum_k (\text{Prop}_k - \bar{\text{Prop}})^2$. λ tunes fairness vs total variance.

How to pick weights / λ in practice

- Cross-validation on downstream task performance with fairness-aware metrics.
- Stakeholder-driven constraints (e.g., minimum acceptable fidelity per group).
- Pareto analysis: compute Pareto frontier of (global explained variance, disparity) to show trade-offs.